Answer **all** the questions in this Section in the spaces provided.

1 During a space expedition on the Moon, a table tennis ball is dropped in a large, enclosed container on the surface of the Moon. The container contains air from the Earth. Special arrangements are made to ensure that the pressure of the air is maintained at the Earth's atmospheric pressure. The variation with time *t* of the speed *v* of the table tennis ball is shown in Fig. 1.1.

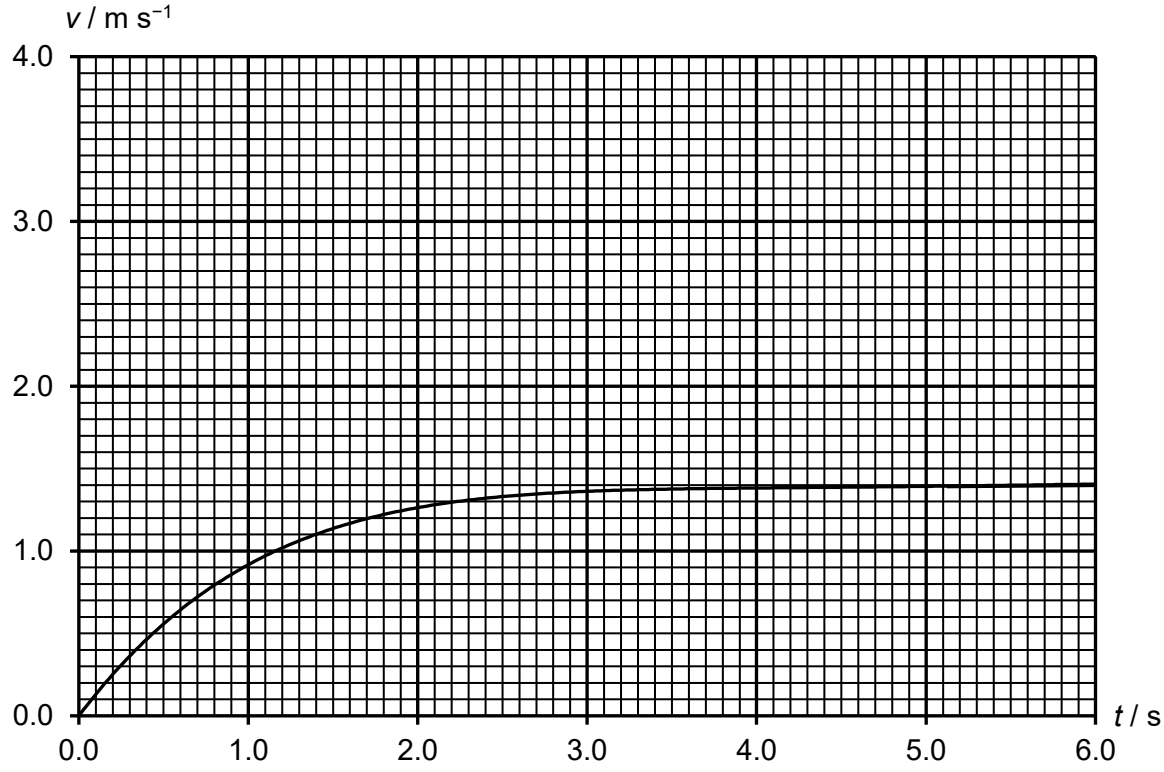

Fig. 1.1

The mass of the table tennis ball is 2.7 g.

(a) **(i)** Use Fig. 1.1 to determine the acceleration of free fall *g* of the ball on the surface of the Moon. Show your construction on Fig. 1.1.

g = ... m s^{-2} [2]

(ii) If the table tennis ball were dropped *outside* the container, determine the time taken for it to reach the value of the constant speed in Fig. 1.1.

time = ... s [1]

(b) The resistive force F acting on the table tennis ball is related to its speed v by the equation

$$F = kv$$

where k is a constant.

(i) Calculate the maximum resistive force experienced by the ball as it falls inside the container.

maximum resistive force = ... N [1]

(ii) Using your answer in **(b)(i)**, deduce the value of k.

k = ... kg s^{-1} [1]

(iii) Determine the maximum speed of the table tennis ball if this experiment is conducted on the surface of the Earth.

maximum speed = ... m s^{-1} [2]

(iv) Sketch, on Fig. 1.1, a new graph showing the variation with time t of the speed v of the table tennis ball when the experiment is repeated with the following changes made independently:

1. Liquid nitrogen of 1.5 times the mass of the ball is injected into the ball. Label this graph P.

2. The ball is thrown vertically downwards with an initial speed of 4.0 m s^{-1}. Label this graph Q.

[3]